

DARPA-SN-25-55

Special Notice

Request for Information: DoD
Relevant System Nominations to
Deliver Resilient Software
Systems

DARPA-SN-25-55

March 10, 2025



Defense Advanced Research Projects Agency
675 North Randolph Street
Arlington, VA 22203-2114

Request for Information (RFI)

DoD Relevant System Nominations to Deliver Resilient Software Systems

Defense Advanced Research Projects Agency (DARPA) Information Innovation Office (I2O)

Posting Date: March 10, 2025

Responses Due: April 03, 2025 at 5:00 PM Eastern Time (ET)

E-mail: DARPA-SN-25-55@darpa.mil

RFI DESCRIPTION

The Defense Advanced Research Projects Agency (DARPA) Information Innovation Office (I2O) seeks information on current and/or future Department of Defense (DoD) systems that would benefit from the use of tools employing software development practices based on formal mathematical methods (i.e., 'formal methods') to verify mathematically that, after application of these tools, systems act in ways that are intended and not in unintentional ways. DARPA is interested in feedback from all capable sources, particularly those sources with formal methods implementation experience.

The goal of this RFI is to solicit information about DoD systems where DARPA tools and expertise can improve the resilience to adversarial attacks.

BACKGROUND

The DoD has been conditioned to accept that software is always vulnerable and that the best available mitigation is to apply retroactive patches as vulnerabilities become known. While patching after exploitation is a damaging strategy for data loss, it is a fatal strategy during active conflict for operational systems. Aligned with the DoD's direction to rebuild our military, we must eliminate acceptance of the status quo identification and patching cycle.

Over the past decade, DARPA has produced a set of scalable, open-source tools that can secure and prove the absence of exploitable vulnerabilities across nearly all DoD systems. These tools employ software development practices based on formal mathematical methods ("formal methods"). For purposes of this RFI, the phrase "formal methods" means techniques used to model aspects of complex systems or subsystems, where mathematical proofs are used to show that the systems will act in ways that are intended and not in unintended ways. The phrase "formal methods" should be broadly interpreted and includes tools such as type-checkers, which complete complex mathematical computations, but hide this complexity from end-users. The broad applicability of these software tools provides a channel for drastically improving the security of DoD systems and ensuring the warfighter can successfully meet complex challenges.

DARPA is working on additional pathways to support the transition of these critical tools through capstone efforts, in which DARPA and military service partners nominate platforms to apply formal methods tools and conduct performance assessments before and after said application to track benefits. Information captured during these capstone efforts will be collected and shared in a Best Practices Guide, as appropriate.

REQUESTED INFORMATION

DARPA is requesting high-level information on DoD systems that would benefit from the use of formal methods as described above. Please see Appendix A for further details. Responses are welcome from all capable sources including, but not limited to, private or public companies, individuals, universities, university-affiliated research centers, not-for-profit research institutions, and U.S. Government-sponsored labs. Note that DARPA assumes that feedback through standard scientific discourse, including direct communication with the corresponding authors or through additional manuscripts, will proceed as usual and views this RFI as in addition to that discourse.

SUBMISSION INSTRUCTIONS

Responses to this RFI should be submitted by April 03, 2025 at 5:00 PM ET.

Unclassified responses to this RFI should be submitted to DARPA-SN-25-55@darpa.mil.
CLASSIFIED INFORMATION SHOULD NOT BE SENT TO DARPA-SN-25-55@darpa.mil.

To the maximum extent possible, respondents should submit non-proprietary information. If proprietary information is submitted, it is the respondent's responsibility to clearly mark what is considered proprietary data. All proprietary information should be clearly labeled "Proprietary." Note, "confidential" is not a classification marking used to control the dissemination of U.S. Government National Security Information as dictated in Executive Order 13526 and should not be used to identify proprietary business information. DARPA will disclose submission contents only for the purpose of review by DARPA staff, other Government agencies, or DARPA Support Contractors/SETAs.

Controlled Unclassified Information (CUI) responses: For unclassified responses containing CUI, applicants will ensure personnel and information systems processing CUI security requirements are in place. If an unclassified submission contains CUI or the suspicion of such, as defined by Executive Order 13556 and 32 CFR Part 2002, the information must be appropriately and conspicuously marked CUI in accordance with DoDI 5200.48. Unclassified submissions containing CUI may be submitted via DARPA-SN-25-55@darpa.mil. Entities generating and submitting CUI responses must do so on an information system authorized to process CUI information IAW NIST SP 800-171 and DoDI 8582.01.

Classified responses: Respondents intending to provide a classified response should send an e-mail notification no later than March 20, 2025 to DARPA-SN-25-55@darpa.mil with the subject line "Classified Submission Requested in order to coordinate the transmission of classified information to DARPA." Classified responses may not be considered if advanced notification is not received by March 20, 2025. **NO CLASSIFIED INFORMATION SHOULD BE SENT TO DARPA-SN-25-55@darpa.mil.**

For classified responses, respondents will ensure all industrial, personnel, and information systems processing security requirements are in place and at the appropriate level (e.g., Facility Clearance Level (FCL), Automated Information Security (AIS), Certification and

Accreditation (C&A), and any Foreign Ownership Control and Influence (FOCI) issues are mitigated prior to classified response generation. Additional information on these subjects can be found at <https://www.dcsa.mil/>.

If a submission contains Classified National Security Information or the suspicion of such, as defined by Executive Order 13526, the information must be appropriately and conspicuously marked with the proposed classification level and declassification date. Submissions requiring DARPA to make a final classification determination shall be marked as follows: “CLASSIFICATION DETERMINATION PENDING. Protect as though classified _____ [insert the recommended classification level, e.g., Confidential, Secret, or Top Secret].”

Respondents choosing to submit classified information from other classified sources (i.e., sources other than DARPA) must (1) Document authorization from the cognizant government program office (e.g., Contracting Officer, Program Manager) to submit classified information in response to this RFI; (2) Ensure the submission is marked in accordance with the source Security Classification Guide (SCG) from which the material is derived; and (3) Include a copy of the source SCG with the RFI response.

Respondents will use transmission, classification, handling, and marking guidance provided by previously issued SCGs, the DoD Information Security Manual (DoDM 5200.01, Volumes 1 - 4), and the National Industrial Security Program Operating Manual, including the Supplement Revision 1 (DoD 5220.22-M and DoD 5200.22-M Sup. 1), when submitting Confidential, Secret, and/or Top Secret classified information. Submissions containing both classified information and CUI must be appropriately and conspicuously marked with the proposed classification level, as well as ensuring CUI is marked in accordance with DoDI 5200.48.

SCI information: Respondents will ensure that SCI information provided is marked, handled, and transmitted in accordance with DoDM 5105.21 Volumes 1-3. Questions regarding the transmission of SCI may be sent to the DARPA Program Security Officer (PSO) via DARPA-SN-25-55@darpa.mil or by contacting the DARPA Special Security Officer (SSO) at 703-812-1970.

SAP information: Respondents will ensure that any SAP information included in responses is marked and handled in accordance with DoDM 5205.07 and transmitted by specifically approved methods, which will be provided in response to the notification of intent to provide a classified response. Respondents choosing to submit SAP information from an agency other than DARPA are required to provide written authorization from the source material’s cognizant Special Access Program Control Officer (SAPCO) or designated representative. For clarification regarding this process, contact DARPA-SN-25-55@darpa.mil or the cognizant DARPA PSO at 571-218-4626.

NOTE: DARPA may reach out to respondents to clarify and/or discuss technical information received against this RFI.

FORMAT AND CONTENT INSTRUCTIONS

DARPA is strongly encouraging all respondents to submit a single integrated response. All submissions must be submitted as an unprotected Microsoft Word document or PDF file. Submissions not meeting this requirement may be discarded.

All submissions must be written in English, and all pages shall be formatted for printing on 8-1/2 by 11-inch paper with 1-inch margins and font size no smaller than 12-point font. Font sizes of 8 or 10-point may be used for figures, tables, and charts. Submissions should be concise and must not exceed 10 pages; submissions in excess of the page limit may not be reviewed.

Page limit includes:	Page limit does NOT include:
Technical feedback	Cover Sheet
Technical figures, tables, charts	References (Optional)

See Appendix A for the RFI submission template, which describes content requirements.

DARPAConnect

Entities who have not worked with DARPA before are encouraged to learn more about DARPAConnect, an initiative established to facilitate collaboration between DARPA and potential performers. The DARPAConnect team offers customized support, resources, and guidance on how to prepare your ideas for high-impact conversations with DARPA program managers. Please visit DARPAConnect.us to access a digital hub of online resources, including a curriculum for self-paced learning, personalized support, and in-person and virtual events. In addition to the self-paced online materials, the DARPAConnect team is able to schedule one-on-one conversations to discuss your specific ideas, questions, and paths to DARPA. You can use the contact form at DARPAConnect.us or email the DARPAConnect team directly at darpaconnect@darpa.mil to request assistance.

ADMINISTRATIVE

This RFI contains all information required to submit a response. No additional forms, kits, or other materials are needed. All administrative and technical questions should be directed to DARPA-SN-25-55@darpa.mil. Please refer to the Special Notice number (DARPA-SN-25-55) in all correspondence.

This RFI does not constitute a formal solicitation for abstracts or proposals; any abstract or proposal received against this RFI will be disregarded. In accordance with FAR 15.201(e), responses to this notice are not offers and cannot be accepted by the Government to form a binding contract. DARPA will not provide reimbursement for costs incurred in responding to this RFI. Respondents are advised that DARPA is under no obligation to acknowledge receipt of the submission received, invite respondents to events, include tools identified in responses, or provide feedback to respondents with respect to any information submitted under this RFI.

APPENDIX A
RFI SUBMISSION TEMPLATE
DARPA-SN-XX-XXXX

COVER SHEET

Submission Title	
Submitter Organization	
Technical Point of Contact (POC)	Name: Address: Telephone: Email:
Administrative POC	Name: Address: Telephone: Email:

[Content requirements are stated in blue font and should be deleted prior to proposal submission. Responses to the RFI should cover at least one of the following questions.]

- 1) Please nominate one or more DoD systems to serve as a Capstone. Nominated systems should be relatively complex systems that have unmet resilience challenges. Each system should be nominated as a separate entry. For each nominated system, please provide the following information:
 - a) The name of the system;
 - b) The entity or entities that have responsibility for the system;
 - c) All government POCs for the system, including technical and administrative;
 - d) A paragraph describing the reasons for nominating a particular system; and
 - e) Particular resilience challenges the system demonstrates.
- 2) Are there specific DoD resilience challenges that are not sufficiently addressed by state-of-the-practice methodologies that might be suitable targets for formal-methods-based tooling? If yes, please explain.